

A Project II Report on

Automated Lemon Leaf Disease Detection Using CNN

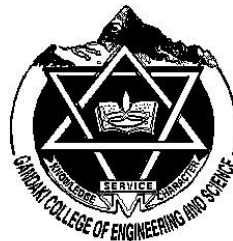
Submitted in partial fulfillment of the requirements for the degree of
Bachelor of Engineering in Software Engineering at Pokhara University

By

Bhawana KC

Ganga Adhikari

Prasamsha Tripathi



Department of Research and Development

GANDAKI COLLEGE OF ENGINEERING AND SCIENCE

Lamachaur, Kaski, Nepal

(August, 2026)

A Project II Report on

Automated Lemon Leaf Disease Detection Using CNN

Submitted in partial fulfillment of the requirements for the degree of
Bachelor of Engineering in Software Engineering at Pokhara University

By

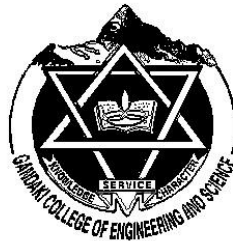
Bhawana KC

Ganga Adhikari

Prasamsha Tripathi

Supervisor

Dr. Bidur Devkota



Department of Research and Development

GANDAKI COLLEGE OF ENGINEERING AND SCIENCE

Lamachaur, Kaski, Nepal

(August, 2025)

BONAFIDE CERTIFICATE

This is to certify that this project titled “**Automated Lemon Leaf Disease Detection using CNN**” in partial fulfillment of the requirements for the degree of BACHELOR OF ENGINEERING IN SOFTWARE ENGINEERING is a bonafide work of **Bhawana KC, Ganga Adhikari** and **Prasamsha Tripathi** under the supervision of **Dr. Bidur Devkota**. It is further certified that this work doesn't form part of any other project work on the basis of which a degree or award was conferred on any earlier occasion on this by any other candidate.

Date of Evaluation: 08/12/2026

.....

Er.

External Examiner

Assistant Professor

School of Engineering,

Pokhara University

.....

Dr. Bidur Devkota

(Supervisor)

.....

Er. Santosh Panth

Project Head

Research Management Cell

ACKNOWLEDGEMENT

In the accomplishment of this project successfully, we want to thank all the people who have contributed their time and effort to guide us throughout the development of the mobile application named 'Automated Lemon Leaf Disease Detection using CNN'.

Primarily, we would like to express our special thanks of gratitude to our teachers and supervisor Dr. Bidur Devkota who guide us to complete this wonderful project on the topic of Automated Lemon Leaf Disease Detection using CNN, which also helped us in doing a lot of research.

Secondly, we would also like to thank Sujata K.C. Technical Officer (Plant Pathology) Nepal Agricultural Research Council, Horticulture Research Station, Malepatan, Kaski, Nepal who guide us in identifying lemon leaf diseases in manually collected lemon leaf data and treatment recommendation for our project

Lastly, parents and friends who helped us a lot in finalizing this project within the limited time frame.

Thank You

ABSTRACT

Lemon cultivation is an important agricultural activity in many regions, contributing significantly to farmers' income and local economies. However, lemon plants are highly susceptible to various diseases such as citrus canker, greening disease, powdery mildew, and fungal infections, which can severely affect crop quality and yield. Early and accurate disease identification is essential for effective disease management, yet traditional diagnosis methods often require expert knowledge that may not be readily accessible to farmers. This project proposes the development of an intelligent lemon leaf disease detection and treatment recommendation system using deep learning techniques. The system will utilize Convolutional Neural Networks (CNNs) to analyze images of lemon leaves and automatically classify them into nine different classes, including healthy and diseased categories. Based on the detected disease, the system will also provide appropriate treatment recommendations to assist farmers in taking timely corrective actions. The proposed solution aims to offer a fast, accurate, and user-friendly tool that can support precision agriculture practices and reduce crop losses caused by delayed disease diagnosis. By leveraging artificial intelligence and image processing technologies, the project seeks to improve disease management and contribute to sustainable lemon farming.

Keyword: *CNN, Deep learning, Citrus Canker, Mildew, Fungal, Precision*

Table of Contents

BONAFIDE CERTIFICATE	iii
ACKNOWLEDGEMENT	iv
ABSTRACT	v
CHAPTER 1 INTRODUCTION	1
1.1. Background.....	1
1.2. Problem Statement.....	1
1.3. Objective.....	2
1.4. Implication.....	2
CHAPTER 2 LITERATURE REVIEW	3
2.1 Comparison Table of Top Performing Models	7
2.2. Research Gap	8
CHAPTER 3 TOOLS AND METHODOLOGY.....	9
3.1. Required Tools	9
3.2. Analysis	11
3.2.1. Use Case Diagram.....	11
3.2.2. System Sequence Diagram.....	13
3.3. Design	15
3.3.1. Proposed Methodology	15
3.4. Approach.....	23

CHAPTER 4 RESULT AND DISCUSSION	27
4.1. Model1 Binary Classification.....	28
4.1.1 Accuracy Curve Analysis.....	28
4.1.2. Loss Curve Analysis	29
4.1.3 Confusion Matrix Analysis	30
4.1.4 Summary	31
4.2 Model 2: Multi-Class Disease Classification	32
4.2.1 Accuracy Curve Analysis.....	32
4.2.2 Loss Curve Analysis	33
4.2.3 Confusion Matrix Analysis	34
4.2.4 Summary	35
CHAPTER:5 CONCLUSION	36
5.1 Conclusion.....	36
5.2 Limitations	36
5.3 Future Work	37
BIBLIOGRAPHY.....	38
APPENDICES.....	39

List of Figures

Figure 1:Use Case Diagram	11
Figure 2:System Sequence Diagram.	13
Figure 3:Proposed Methodology.....	15
Figure 4:Architecture of Convolutional Neural Network (CNN).....	23
Figure 5:Architecture of MobileNetV2.....	26
Figure 6:Binary Classification Accuracy Curve.....	28
Figure 7:Binary Classification Loss Curve	29
Figure 8:Binary Classification Confusion Matrix.....	30
Figure 9: Disease Classification accuracy Curve.....	32
Figure 10: Disease Classification Loss Curve	33
Figure 11: Disease Classification Confusion Matrix	34

List of Tables

Table 1:Comparison of past works	7
Table 2: Dataset.....	19
Table 3:Binary Classification Report.....	22
Table 4: Multiclass Disease Classification	22

CHAPTER 1

INTRODUCTION

1.1. Background

Agriculture plays a vital role in the economy of Nepal, providing livelihood to a large portion of the population. However, plant leaf diseases significantly reduce crop productivity and quality, especially in rural and hilly regions where access to agricultural experts is limited. Farmers mainly rely on manual visual inspection to identify diseases, which is often time-consuming, inconsistent, and prone to error. Early detection is difficult using traditional methods, leading to crop damage and economic loss. With advancements in Artificial Intelligence and deep learning, image-based disease detection systems have become effective solutions for accurate and real-time diagnosis. Therefore, the proposed Leaf Disease Detection System aims to provide an accessible, AI-based tool that helps farmers identify disease early and improve crop management

1.2. Problem Statement

In recent years, citrus orchards in our locality have experienced progressive decline, leading to significant crop losses, reduced agricultural productivity, and negative impacts on smallholder farmers' livelihoods. Although the region is agriculturally important, the early identification and systematic analysis of lemon leaf diseases remain limited due to the lack of accessible and technology-based diagnostic tools. Therefore, there is a need to develop an automated CNN-based system that can accurately detect and classify lemon leaf diseases from images to support early and reliable diagnosis.

1.3. Objective

- To provide farmers with an easy-to-use interface that allows them to either capture a live photo of a lemon leaf or upload an existing image from their device for leaf classification and treatment recommendations.

1.4. Implication

The CNN-based deep learning system for lemon leaf disease detection and treatment suggestion has strong implications across several areas. It enables farmers to quickly identify plant diseases through simple image input, allowing timely action before damage spreads. Automated treatment suggestions provide instant, accurate guidance, removing the need for costly expert consultation. Economically, early detection reduces crop losses and protects farmer income by preventing unnecessary spending on wrong treatments. The disease data generated by the system helps agricultural authorities plan better intervention programs and monitoring systems. Environmentally, precise treatment guidance reduces overuse of pesticides, promoting safer and more sustainable farming. The framework developed can be replicated and applied to other crops and regions, making it academically valuable. Socially, the system brings expert-level agricultural knowledge directly to rural farmers, improving their confidence and decision-making. Overall, this research connects artificial intelligence with practical farming needs, offering a simple, scalable, and effective solution for disease management in citrus growing regions.

CHAPTER 2

LITERATURE REVIEW

Arifin et al. (2024) proposed a hybrid disease classification system for citrus fruits specifically lemons and oranges by combining deep learning feature extraction with traditional machine learning classifiers. The study employed three pretrained convolutional neural network (CNN) architectures, namely VGG16, VGG19, and ResNet50, exclusively for feature extraction, followed by classification using Random Forest, Naive Bayes, K-Nearest Neighbors (KNN), and Logistic Regression. Datasets consisting of 200 lemon images across four disease classes (canker, mold, scab, and healthy) and 1,614 orange images across four classes (blackspot, canker, fresh, and greening) were sourced from Kaggle. Experimental results demonstrated that the combination of ResNet50 as a feature extractor and Logistic Regression as a classifier achieved the highest accuracy of 95.0% for lemon disease classification and 99.69% for orange disease classification. The proposed hybrid approach outperformed standalone CNN models such as VGG16 and ResNet50 used directly with Softmax classifiers, highlighting the effectiveness of integrating deep learning feature representations with classical machine learning algorithms for agricultural disease detection (Arifin & Jahan, 2024).

Saygılı (2025) proposed a hybrid classification framework for detecting diseases in citrus leaves by integrating transfer learning-based deep learning models with classical machine learning techniques. The study employed four pretrained CNN architectures DenseNet201, ResNet50, MobileNetV2, and EfficientNet-B0 for feature extraction, which were subsequently refined using the minimum redundancy maximum relevance (mRMR) feature selection method to eliminate redundant information while retaining the most discriminative attributes. The selected features were then classified using support vector machines (SVM), ensemble bagged trees, k-nearest neighbors (KNN), and neural networks under stratified 10-fold cross-validation to ensure statistical reliability and prevent data leakage. Experiments were conducted on a lemon leaf dataset comprising 23 images sourced from the Kaggle

platform, with additional cross-domain validation performed on mango and pomegranate leaf datasets. Results demonstrated that the combination of DenseNet201 and SVM without feature selection achieved the highest classification performance, reaching 94.1% accuracy and 93.2% F1 score for lemon disease detection. The same pipeline achieved 100.0% accuracy on mango and 98.7% on pomegranate datasets, confirming strong cross-species generalization. Furthermore, lightweight architectures such as EfficientNet-B0 and MobileNetV2 offered significantly higher inference throughput, making them suitable candidates for real-time field deployment in resource-constrained agricultural environments (Saygılı, 2025).

Dhiman et al. (2023) proposed a smart citrus fruit disease detection system by integrating a hybrid CNN-LSTM deep learning model with edge computing technology. The study addressed a critical limitation of conventional deep learning models their high computational demands make them unsuitable for deployment on resource-constrained edge devices used in real agricultural settings. To overcome this, the researchers incorporated magnitude-based pruning and post-quantization techniques to compress the model while maintaining classification accuracy. The dataset comprised 2,950 citrus fruit images covering six categories blackspot, canker, scab, melanosis, greening, and healthy collected from Kaggle repository. Image preprocessing involved rescaling to 256×256 pixels, data augmentation expanding the dataset to 20,448 samples, K-means clustering for segmentation, and Canny edge detection. Experimental results demonstrated that the CNN-LSTM model without compression achieved 98.87% accuracy, while the magnitude-based pruning variant reduced model size by 50.29% with only a modest accuracy trade-off. The combination of pruning with post-quantization further reduced model size, making it deployable on low-resource edge devices. The proposed model consistently outperformed the baseline CNN approach across precision, recall, and F-score metrics, highlighting the practical advantage of combining deep learning with model compression strategies for real-time agricultural disease detection (Dhiman & Ababneh, 2023).

Da Silva et al. (2023) proposed a comprehensive mobile edge AI pipeline for detecting and mapping citrus fruit diseases directly within orchards, addressing the practical limitation of cloud dependency in field environments where internet connectivity is unreliable. The study followed a three-stage approach first detecting citrus fruits in orchard images, then classifying detected fruits as healthy or diseased, and finally mapping the spatial distribution of diseases across the orchard area. For the fruit detection task, YOLO-V3 and Faster R-CNN were compared, with YOLO-V3 demonstrating a significantly superior processing speed-achieving detection in under 100 milliseconds-despite both models yielding similar mean average precision scores of 88% and 90% respectively. For disease classification, three lightweight CNN architectures-MobileNetV2, EfficientNetV2-B0, and NASNet-Mobile were evaluated on a dataset covering four citrus conditions: canker, fresh, greening, and blackspot. MobileNetV2 and EfficientNetV2-B0 both achieved 100% classification accuracy, while NASNet-Mobile reached 98%. MobileNetV2 was identified as the optimal model due to its 10% timing advantage over EfficientNetV2-B0, making it the most suitable candidate for real-time mobile deployment. Additionally, the study introduced a novel disease spread mapping method using a bivariate normal probability function combined with a genetic algorithm for parameter regression, enabling spatial visualization of disease distribution across a simulated 1-hectare orchard. The proposed pipeline demonstrated that edge-based AI can effectively support precision agriculture by providing accurate, real-time, and deployable disease detection without reliance on cloud infrastructure (da Silva & Oliveira, 2023).

Pramanik et al. (2021) proposed a CNN-based transfer learning system for classifying lemon leaf diseases into three categories citrus disease, pest disease, and no disease. The study evaluated four pretrained deep learning architectures, namely DenseNet-201, ResNet-50, ResNet-152V2, and Xception, using a real-life dataset of 314 lemon leaf images collected directly from nurseries and gardens in Dhaka, Bangladesh. The dataset was carefully verified by taking feedback from 10 people per image and consulting plant disease experts, ensuring reliable labeling. For preprocessing, unclear and unnecessary images were removed, and the remaining data was resized, labeled, and split into training and test sets with an 83:17 ratio 261 training images and 53 test

images. All four models were trained using pre-trained ImageNet weights with transfer learning, and two dropout layers with a rate of 0.5 were applied after the fully connected layer to reduce overfitting. Experimental results demonstrated that the exception model outperformed all other architectures, achieving the highest overall accuracy of 94.34%, along with a micro-average AUC of 0.995 and macro-average AUC of 0.996 on the ROC curve. DenseNet-201 followed with 92.45% accuracy, ResNet-50 achieved 88.68%, and ResNet-152V2 performed lowest at 79.25%. The study also outperformed several existing citrus disease detections works including GLCM-based methods (85.71%) and earlier CNN-SVM hybrid approaches (93.8%). The authors acknowledged that the number of disease classes could be increased in future work to make the system more comprehensive (Aktaruzzaman Pramanik, 2021).

Kalejahi et al., 2026. research systematically evaluates the performance of MobileNetV2 and ResNet50 architectures for early plant disease detection using the New Plant Disease dataset, which comprises approximately 87,000 images spanning 38 classes, with an emphasis on developing scalable solutions suited to resource-constrained environments such as drones, smartphones, and IoT systems rather than pursuing accuracy alone. The best-performing configuration employed the Adam optimizer with an initial learning rate of $1e-4$, trained over 25 epochs with a batch size of 32, and combined Dropout (0.5) with L2 regularization ($1e-4$), alongside data augmentation techniques including random rotations of $\pm 25^\circ$, flips, and 20% shifts in width and height. Under this setup, MobileNetV2 achieved a superior accuracy of 98.33% while requiring seven times fewer parameters (approximately 3.5M compared to ResNet50's 25M) and cutting training time per epoch by roughly 40%, underscoring its efficiency advantage for lightweight deployment. Building on these findings, future work will focus on validating the model under real-world field conditions, integrating multispectral imaging data to enrich detection capability, and incorporating Explainable AI (XAI) techniques to strengthen user trust and interpretability of the system's predictions (Kalejahi, 2026).

2.1 Comparison Table of Top Performing Models

Table 1: Comparison of past works

SN	Author (Year)	Best Model Architecture	Key Hyperparameters	Accuracy (Lemon)	Future Work Focus
1.	B. K. Kalejahi et al. (2026)	MobileNetV2	Adam, LR: 1e-4, Batch: 32, Dropout: 0.5	98.33%	Field validation, XAI integration
2.	Saygılı (2025)	DenseNet201 + SVM	RBF Kernel, C: 2.0, mRMR k: 256	94.1%	Distillation, broader citrus cultivars
3.	Arifin et al. (2024)	ResNet50 + Logistic Regression	RGB images, L1/L2 Regularization	95.0%	Original field datasets, severity levels
4.	Dhiman et al. (2023)	CNN-LSTM	SGD Optimizer, K-means k: 4, Pruning	98.25%	Real-time agricultural image testing
5.	Da Silva et al. (2023)	MobileNetV2 (Classification)	Adam, 512 Dense Neurons, Softmax	100.0%	Temporal dynamics, risk management
6.	Pramanik et al. (2021)	Xception	100 Epochs, Batch: 8, Dropout: 0.5	94.34%	Increasing disease class count

2.2. Research Gap

Existing studies on lemon leaf disease detection are limited by narrow disease categorization, typically covering only two to four classes (e.g., healthy vs. diseased, or basic categories like canker, mold, and scab), which falls short of the comprehensive symptom differentiation needed for detailed field monitoring. While models such as DenseNet201 + SVM or ResNet50 + Logistic Regression report high accuracies of 94.1% to 95% on these simplified lemon datasets, this performance is largely untested in more complex multi-class environments, revealing a tendency to prioritize accuracy on narrow datasets over the broader disease coverage required for real-world agricultural use. Additionally, most high-performing lightweight models are benchmarked on general datasets like the 38-category New Plant Disease dataset rather than specialized, high-diversity lemon datasets, leaving a gap that a granular, 9-class IEEE lemon dataset is better positioned to address.

Research Summary: The broader literature establishes MobileNetV2 as a leading choice for mobile edge AI, offering a superior efficiency-performance trade-off, achieving up to 98.33% accuracy on general plant datasets and even 100% on specific citrus classification tasks, while remaining roughly seven times smaller than ResNet50, making it well-suited for deployment on drones and smartphones. This project directly addresses the identified research gap by expanding classification to 9 distinct lemon leaf classes using MobileNetV2 on a specialized IEEE dataset, achieving 99% accuracy for binary lemon detection and 90% for multi-class disease classification.

CHAPTER 3

TOOLS AND METHODOLOGY

3.1. Required Tools

The following tools are used for development of this project:

- **For backend**

Django: A high-level Python web framework used for building the backend, handling server-side logic, APIs, and database management. It provides a secure and scalable foundation for integrating the trained deep learning model into a web application.

- **For UML diagrams and wireframes**

Draw.io: A free, web-based diagramming tool used to create UML diagrams and wireframes for visualizing system architecture and user interfaces.

- **Deep Learning**

TensorFlow: An open-source deep learning framework used to build, train, and evaluate the MobileNetV2 and ConvnextNet models for plant disease classification.

- **For Gantt chart**

MS-Excel: A spreadsheet application used to create Gantt charts for visualizing the project timeline, tasks, and milestones. It helps track progress and manage deadlines throughout the project lifecycle.

- **For version control and collaboration**

Git and GitHub: Git is a version control system used to track changes in the codebase, while GitHub is a cloud-based platform for hosting repositories and

enabling team collaboration. Together, they ensure smooth code management and easy rollback in case of errors.

- **For coding**

Visual Studio Code: A lightweight yet powerful source-code editor used for writing and debugging code across the project. It supports multiple programming languages and extensions that streamline the development workflow.

- **For dataset**

Kaggle: An online platform used to source the New Plant Disease dataset and the specialized IEEE lemon leaf dataset for model training. It also provides free GPU resources for experimentation and benchmarking. A lightweight yet powerful source-code editor used for writing and debugging code across the project. It supports multiple programming languages and extensions that streamline the development workflow.

- **Training Environment:**

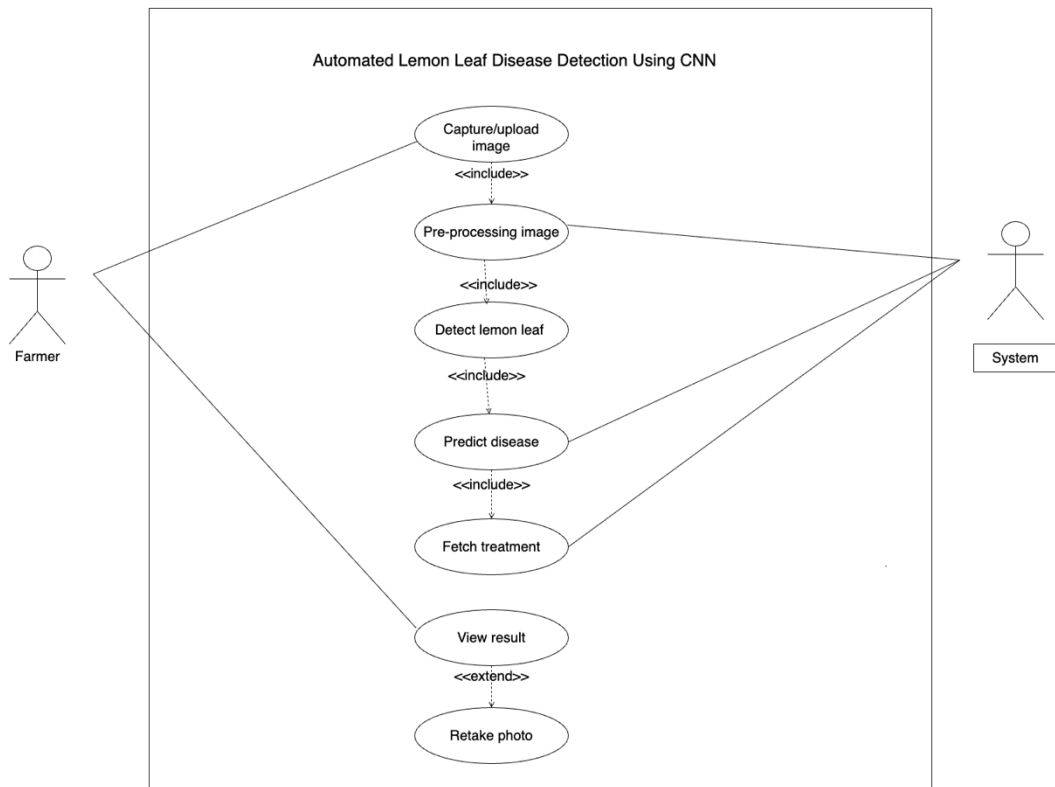
GoogleColab/JupyterNotebook: Cloud-based and local coding environments, respectively, used for training deep learning models with GPU/TPU acceleration. They allow for interactive development, easy visualization of results, and efficient experimentation with hyperparameters.

- **Mobile App**

Flutter (Dart): A cross-platform mobile app development framework using the Dart programming language, used to build the mobile application for end-users. It enables a single codebase to run smoothly on both Android and iOS devices, ensuring wider accessibility for farmers and field workers

3.2. Analysis

3.2.1. Use Case Diagram



(UseCase Diagram)

Figure 1:Use Case Diagram

As showed in [7] the usecase shows the overall actor system relationship

Use Case UC-01: Capture/Upload Image

Precondition: The actor has the application installed and is logged in. A camera or image file presented available on the device.

Postcondition: A valid leaf image successfully submitted and available in the system for further processing.

Use Case UC-02: Pre-processing Image

Precondition: A valid image has been captured or uploaded by the actor (UC-01 must be completed).

Postcondition: The image is resized, normalized, and cleaned ready to be fed into the disease detection model.

Use Case UC-03: Detect Lemon Disease

Precondition: The image has been successfully pre-processed (UC-02 must be completed). The trained ML model is loaded and available.

Postcondition: The system produces a classification result either "Healthy" or a specific disease name with a confidence score.

Use Case UC-04: View Diagnosis and Treatment

Precondition: A disease has been detected and a classification result exists (UC-03 must be completed).

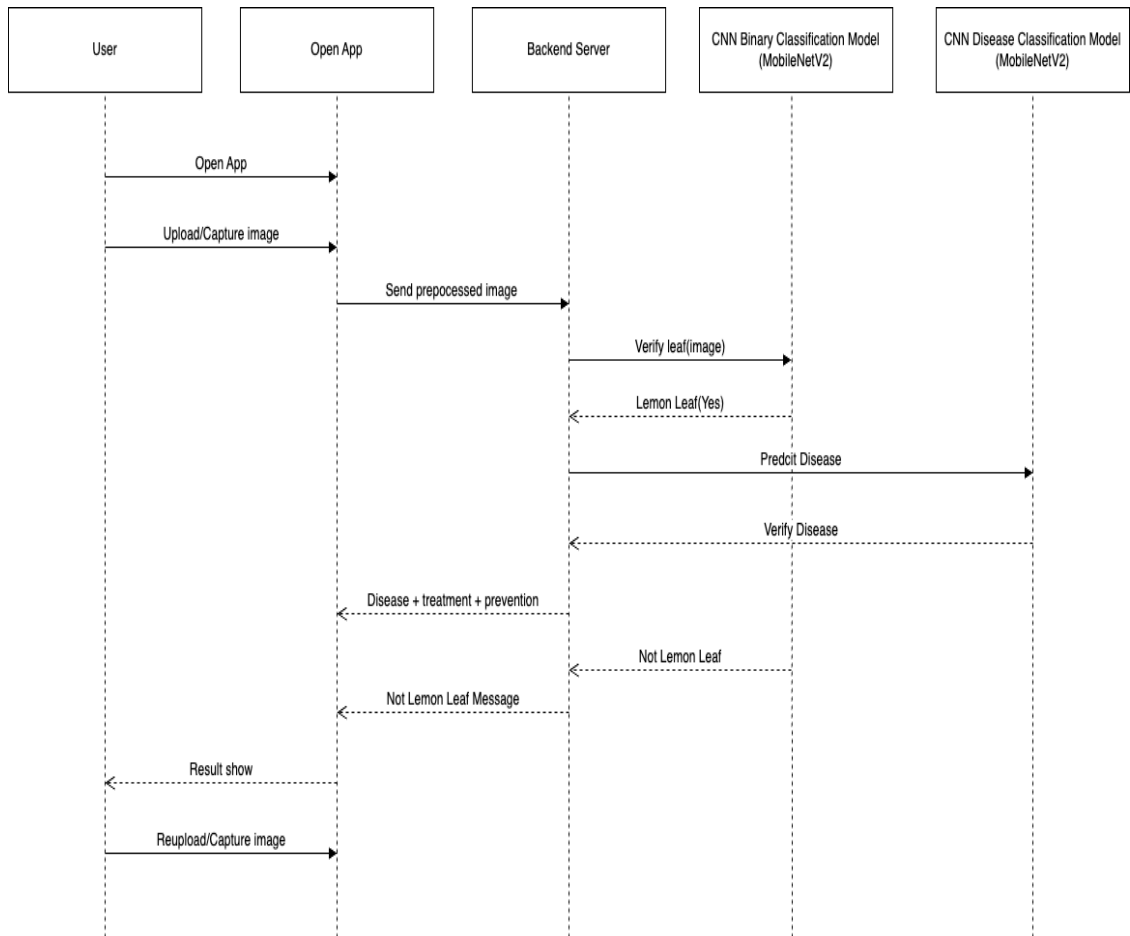
Postcondition: The farmer is presented with the disease name, severity level, and a recommended treatment or prevention plan.

UC-05: Manage and Retrain Model

Precondition: Sufficient new labeled training data is available. The system or admin has access to the model management module.

Postcondition: The AI model is retrained, validated, and the updated version is deployed for future disease detections.

3.2.2. System Sequence Diagram



(SSD)

Figure 2: System Sequence Diagram.

The system works as follows: the user first opens the app and uploads or captures a leaf image, which the app preprocesses and sends to the backend server. The backend then forwards the image to the CNN Binary Classification Model to verify whether it is actually a lemon leaf. If the model confirms it is a lemon leaf, the backend passes the image on to the CNN Disease Classification Model, which predicts the disease and sends the verified result back to the backend. The backend then compiles the disease information along with its treatment and prevention details and sends this combined result back to the app, which displays it to the user. However, if the binary classification model determines that the image is not a lemon leaf, the backend skips

the disease prediction step entirely and instead sends a "Not Lemon Leaf" message back to the app, which is then shown to the user. In either case, the user has the option to reupload or capture a new image to repeat the entire verification and diagnosis process

3.3. Design

3.3.1. Proposed Methodology



Figure 3:Proposed Methodology

Steps of the Proposed Approach

1. Data Collection

A dataset comprising both healthy and diseased lemon leaf images was utilized in this study. The primary dataset is collected from the IEEE research paper dataset, which contains nine classes of lemon leaf conditions: Anthracnose, Bacterial Blight, Citrus Canker, Curl Virus, Deficiency Leaf, Dry Leaf, Healthy Leaf, Sooty Mould, and Spider Mites. Each class consists of 100 images with a resolution of 1000×1000 pixels, resulting in a total of 900 images (Benjamin Doh (Jiangsu University)).

Additionally, another lemon leaf dataset was obtained from the research paper titled “Lemon Leaf Dataset: Images of Citrus Diseases and Healthy Leaves from Bangladesh”. This dataset contains seven classes: Anthracnose (609 images), Citrus Canker (623 images), Citrus Greening (Huanglongbing) (618 images), Citrus Leaf Curl (634 images), Citrus Leaf miner (650 images), Healthy Lemon Leaf (630 images), and Swallowtail Larval Herbivory (Deficiency Leaf) (630 images). All images in this dataset have a resolution of 512×512 pixels (Md Naimul Hasan, 2025).

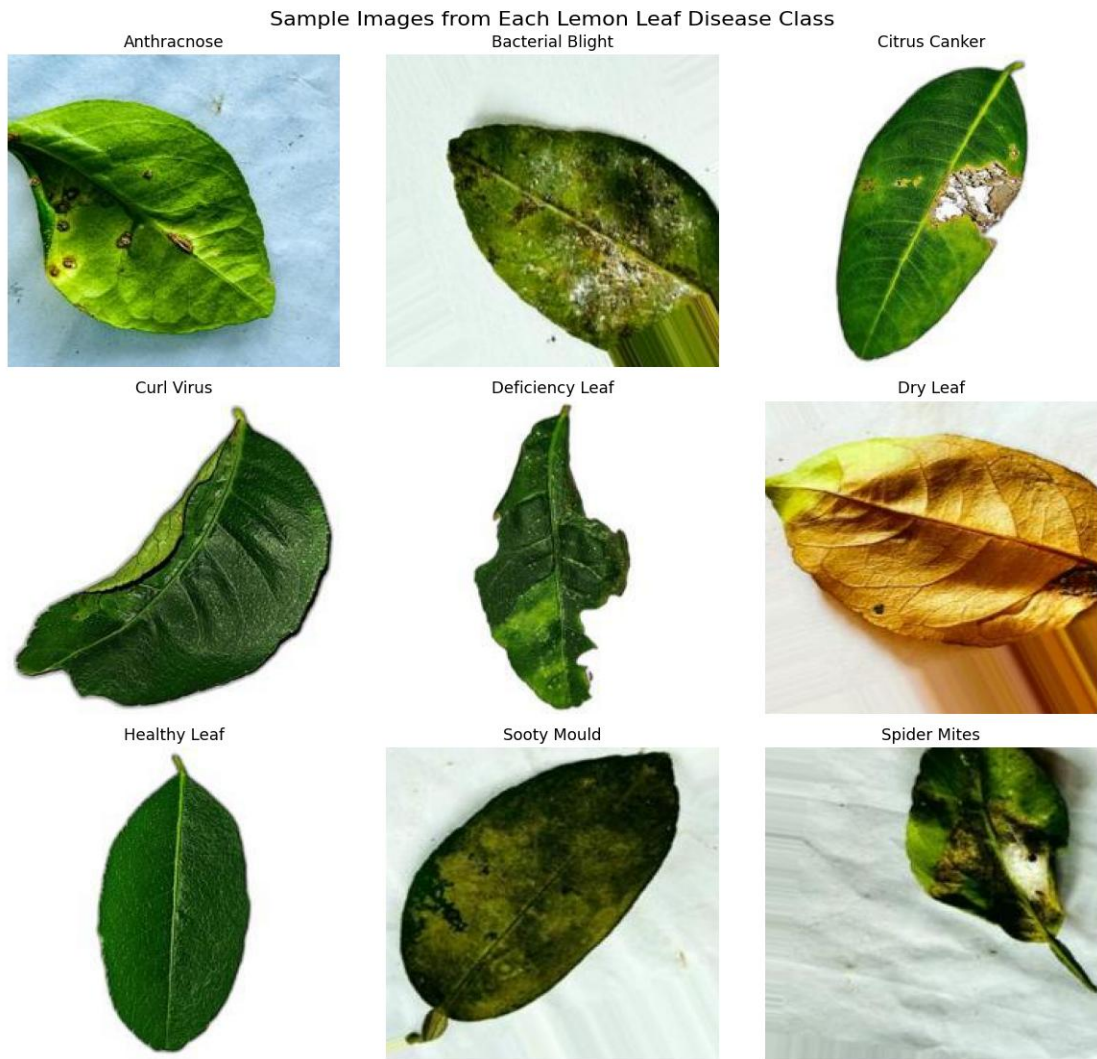


Figure 4: Samples of Lemon Leaf

The two datasets were combined and carefully analyzed to identify common disease categories suitable for model development. From the combined dataset, five classes Anthracnose, Citrus Canker, Curl Virus (Citrus Leaf Curl), Deficiency Leaf (Swallowtail Larval Herbivory (Deficiency Leaf)) and Healthy Leaf were selected for model training and evaluation.

These classes were chosen because they were either common across both datasets or represented the most significant disease conditions affecting lemon leaves. Furthermore, selecting these classes helped maintain class balance and improve the overall learning performance and classification accuracy of the model.

To enhance the robustness of the proposed system, a separate non-lemon leaf dataset was also collected from publicly available research sources and Kaggle. This dataset contains images of leaves from various plant species as well as non-leaf objects.

Sample Images from Not Lemon Leaf Dataset



Figure 5: Samples of Non Lemon Leaf

It was used to develop a preliminary lemon leaf verification model that determines whether an uploaded image belongs to a lemon leaf before disease classification is performed. This additional verification stage helps reduce incorrect predictions and improves the reliability of the overall disease detection system.

2.Data Preprocessing

Data preprocessing for this project began with collecting a lemon leaf image dataset along with a supplementary non-lemon dataset sourced from Kaggle, which was used to enable the model to distinguish lemon leaves from non-lemon images before proceeding to disease classification.

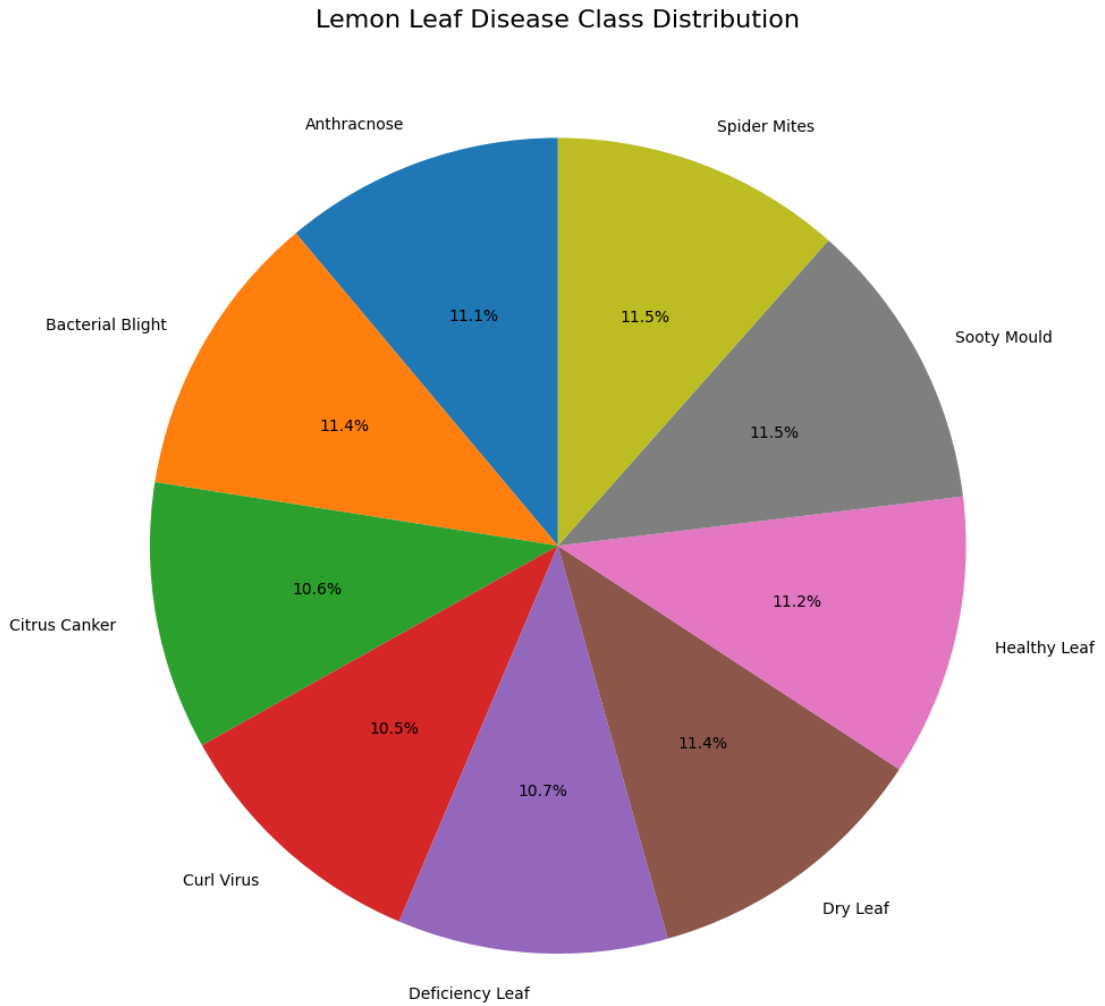


Figure 6: Piechart of Lemon leaf Class Distribution

The combined dataset was organized into class-wise folders and split into training, validation, and test sets to ensure proper model evaluation. Images were then cleaned by removing corrupted, duplicate, and irrelevant samples, followed by resizing all images to a uniform size of 224x224 pixels to match the input requirements of the MobileNetV2 architecture. It is important to note that only the lemon leaf disease dataset underwent data augmentation techniques such as random rotations ($\pm 25^\circ$), horizontal and vertical flips, width/height shifts (20%), and zoom adjustments to increase dataset diversity and reduce overfitting, while the non-lemon dataset was resized only, without augmentation, since it was solely used for the binary lemon vs. non-lemon classification stage. Pixel values across both datasets were normalized to a range of 0 to 1 to stabilize and accelerate the training process. Class labels were encoded using sparse integer encoding paired with the

sparse_categorical_crossentropy loss function, eliminating the need for one-hot encoding while maintaining equivalent classification performance. Finally, the preprocessed images were organized into batches of size 32 to enable efficient and memory-optimized training of the deep learning models.

Table 2: Dataset

SN	Class Name	IEEE Dataset	Research Paper	Data Augmentation	Total
1	Anthracnose	100	609	251	960
2	Bacterial Blight	100	-	885	985
3	Citrus Canker	100	623	196	919
4	Curl Virus	100	634	174	908
5	Deficiency Leaf	100	630	195	925
6	Dry Leaf	100	-	888	988
7	Healthy Leaf	100	630	235	965
8	Sooty Mould	100	-	897	997
9	Spider Mites	100	-	897	997
	Total	900	3,126	4,618	8,644

|

3.Model Training

The dataset was split into 64% training, 16% validation, and 20% testing sets to ensure a robust evaluation of model performance. MobileNetV2 was selected as the base architecture for both the binary (lemon vs. non-lemon) and multi-class (disease classification) models due to its lightweight structure and efficiency, making it suitable for deployment on resource-constrained devices. The hyperparameter configuration was adapted from a reference research paper, using the Adam optimizer with a learning rate of 0.0001, a batch size of 32, and a maximum of 25 epochs, with EarlyStopping applied to halt training once validation performance stopped improving, thereby preventing unnecessary computation and overfitting. For the binary classification model (lemon vs. non-lemon), Dropout regularization was removed while retaining all other hyperparameters as specified in the reference paper, since the simpler binary task required less aggressive regularization to avoid underfitting. For the multi-class disease classification model, the full regularization scheme combining Dropout and L2 regularization was applied exactly as outlined in the reference paper, given the increased complexity of differentiating between multiple disease classes. During training, the model learned by minimizing a Cross-Entropy loss function, with weights updated iteratively through the Adam optimizer across epochs to improve classification accuracy. Validation data was used throughout training to monitor performance and detect signs of overfitting in real time. After training, both models were evaluated on the testing dataset using metrics such as accuracy, precision, recall, and confusion matrix to ensure reliable performance before deployment.

4.Evaluation

The performance of the proposed CNN models was evaluated using the following metrics:

1. Accuracy: Accuracy measured the overall correctness of the model by calculating the percentage of correctly classified images out of the total images, showing how well the system predicted both healthy and diseased lemon leaves. The binary lemon vs. non-lemon model achieved a test accuracy of 99.32%, while the multi-class disease classification model achieved a test accuracy of 94.04%.

2. Precision: Precision measured how many of the leaves predicted as diseased were actually diseased, helping reduce false positive results and ensuring reliable disease

detection. The binary model recorded a precision of 0.9964 for the not-lemon-leaf class and 0.9896 for the lemon-leaf class, while the multi-class model achieved a weighted precision of 0.9412 and a macro precision of 0.9394.

3. Recall (Sensitivity): Recall measured how many actual diseased leaves were correctly identified by the model, ensuring that diseased leaves were not missed. The binary model achieved a recall of 0.9907 for the not-lemon-leaf class and 0.9959 for the lemon-leaf class, while the multi-class model recorded a weighted recall of 0.9404 and a macro recall of 0.9387.

4. F1-Score: The F1-score, calculated as the harmonic mean of precision and recall, provided a single balanced metric that accounted for both false positives and false negatives. The binary model achieved an F1-score of 0.9936 for the not-lemon-leaf class and 0.9928 for the lemon-leaf class, while the multi-class model recorded a weighted F1-score of 0.9402 and a macro F1-score of 0.9384.

5. Confusion Matrix: The confusion matrix showed the number of correct and incorrect predictions for each class, helping analyze model performance in detail by comparing predicted and actual values. For the binary model, out of 3,673 total test samples, 3,648 were correctly predicted and 25 were incorrectly predicted, confirming the strong reliability of the lemon vs. non-lemon classification.

6. Loss Function: The loss function measured the difference between predicted results and actual labels during training, where a lower loss value indicated better model performance. The multi-class disease classification model achieved a final training loss of 0.2927, a final validation loss of 0.2056, and a test loss of 0.2151, with the best validation loss of 0.2056 recorded at epoch 25, reflecting stable convergence during training.

Table 3: Binary Classification Report

Metric (Weighted)	Value
Precision	0.99
Recall	0.99
F1-Score	0.99
Average Accuracy	0.9954

Model 1 classification report table

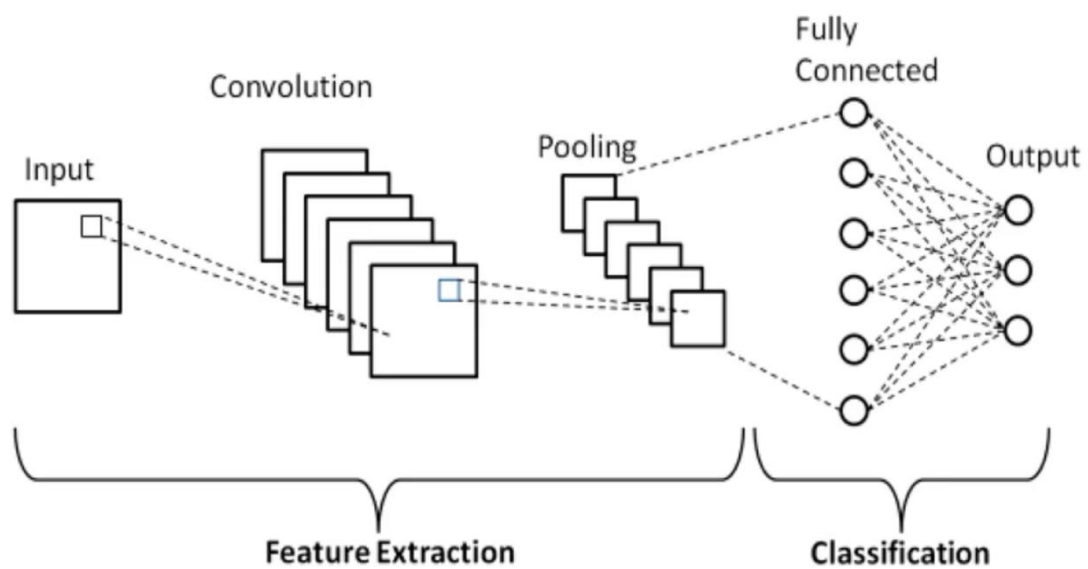
Table 4: Multiclass Disease Classification

Metric (Weighted)	Value
Precision	0.9412
Recall	0.9404
F1-Score	0.9402
Average Accuracy	0.9323

Model 2 classification report table

3.4. Approach

The model system uses a **Convolutional Neural Network (CNN)** for lemon leaf disease classification. CNN is a deep learning algorithm mainly used for image recognition and classification tasks. In this approach, the input leaf image is first passed through convolution layers, where important visual features such as edges, textures, and disease patterns are automatically extracted. Convolution layer applies 32 filters of size 3x3 to the input images with a valid padding. The input shape is (64, 64, 3), indicating that the input images are 64x64 pixels with 3 color channels (RGB).



(CNN)

Figure 7:Architecture of Convolutional Neural Network (CNN)

Steps:

1.Input Layer

The input layer receives the lemon leaf image as the starting point of the network. The image is resized into a fixed dimension (for example, 224×224 pixels) and passed into the model for processing. This layer does not perform any computation; it only feeds the image data into the CNN.

2.Convolution Layer (Feature Extraction)

The convolution layer is the core part of CNN. It applies multiple filters (kernels) to the input image to extract important features such as edges, shapes, textures, and disease spots. These filters help the model detect patterns that represent healthy or diseased leaves. This layer is responsible for learning visual features automatically.

3.Pooling Layer

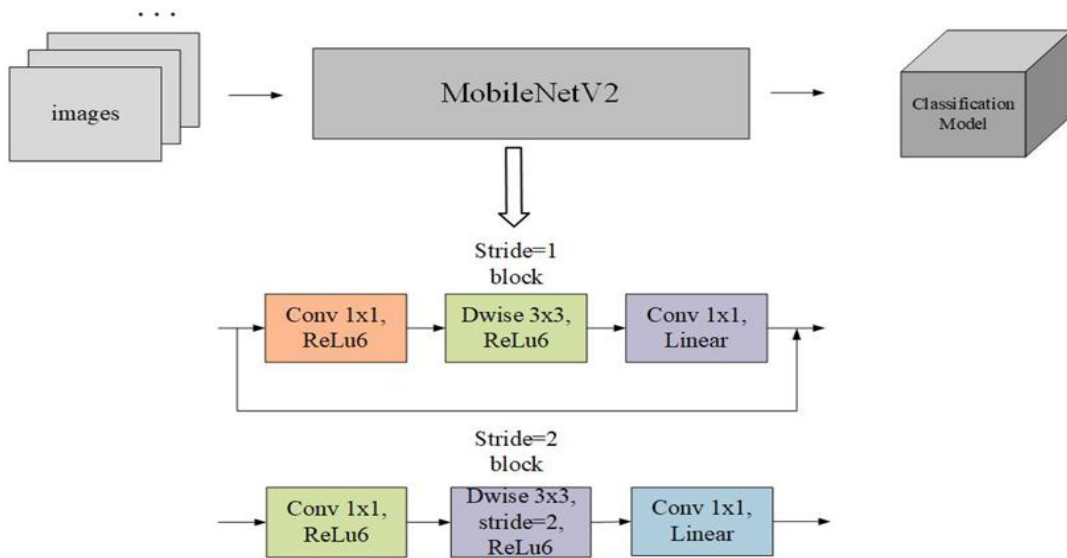
The pooling layer reduces the size of the feature maps generated by the convolution layer. It helps to decrease computational complexity and prevent overfitting. Max Pooling is commonly used, which selects the maximum value from a small region of the feature map. This keeps the most important information while reducing data size.

4.Fully Connected (Dense) Layer

After feature extraction and pooling, the data is flattened into a one-dimensional vector and passed to the fully connected layer. This layer connects all neurons and performs high-level reasoning based on the extracted features. It helps the model learn relationships between features for classification.

5.Output Layer

The output layer produces the final prediction. It uses the activation function to generate probability scores for each class (for example, healthy or diseased). The class with the highest probability is selected as the final result



(MobilenetV2)

Figure 8:Architecture of MobileNetV2

MobileNetV2 is a lightweight and efficient Convolutional Neural Network model that can be used for image classification through transfer learning. In this project, MobileNetV2 will be used to classify lemon leaf disease by utilizing its pre-trained feature and adding custom classification layers for the specific dataset. Since it requires less computational power and provides high accuracy, it is suitable for developing a fast and reliable disease detection system that can be deployed in web or mobile applications.

CHAPTER 4

RESULT AND DISCUSSION

This project produced a trained and optimized CNN model capable of accurately detecting lemon leaf diseases from digital images and generating appropriate treatment suggestions. The outcome included a simple, user-friendly application that allowed farmers with minimal technical knowledge to capture leaf images and instantly receive disease diagnosis and treatment guidance. The model achieved high classification accuracy through proper training and validation on a well-prepared lemon leaf image dataset. Practically, the system contributed to early disease detection, reduced crop losses, and improved farmer income by enabling timely and correct treatment decisions. The research also produced a reusable dataset and a replicable deep learning framework applicable to other crops and regions, contributing meaningfully to the field of precision agriculture.

4.1. Model1: Binary Classification

4.1.1 Accuracy Curve Analysis

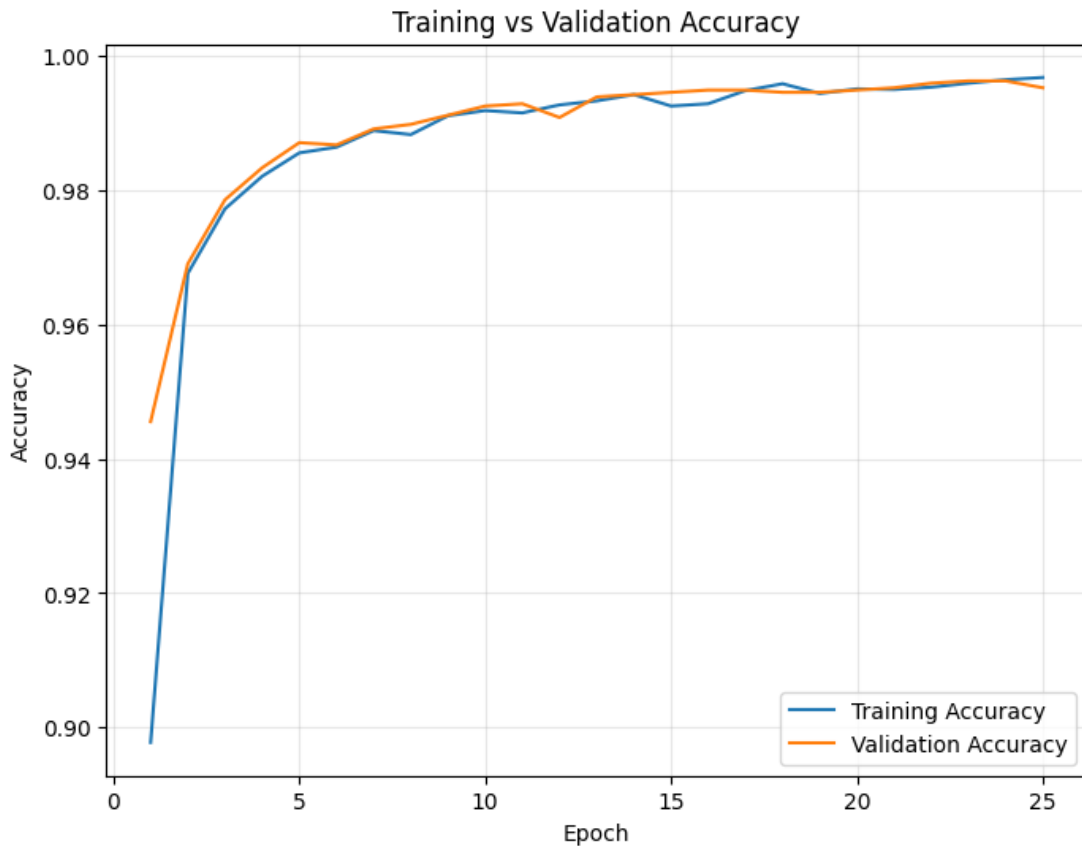


Figure 9: Binary Classification Accuracy Curve

Figure 6 illustrates the training and validation accuracy of the binary lemon-leaf verification model across 25 training epochs. At the start of training (epoch 1), the training accuracy was recorded at approximately 0.898, while the validation accuracy began at a comparatively higher value of approximately 0.945. This initial gap is a normal characteristic of CNN training, arising because regularization mechanisms behave differently during the training phase compared to the evaluation phase.

Both curves exhibited a sharp increase between epoch 1 and epoch 3, rising from the 0.90–0.94 range to approximately 0.97–0.98, indicating that the model rapidly learned the primary distinguishing features between lemon and non-lemon leaf images within the first few epochs. Beyond epoch 5, the rate of improvement slowed considerably, and both curves entered a gradual plateau phase. By epoch 25, the training accuracy reached approximately 0.997, while the validation accuracy stabilized at

approximately 0.995. A minor, brief dip in validation accuracy was observed around epoch 11–12, where it decreased to approximately 0.99 before recovering in the subsequent epoch; this fluctuation is attributed to normal batch-level variance rather than a genuine performance issue.

Throughout the entire training process, the training and validation accuracy curves remained closely aligned, with no widening gap between them. This behavior indicates that the model did not suffer from overfitting and generalized effectively to unseen validation data.

4.1.2. Loss Curve Analysis

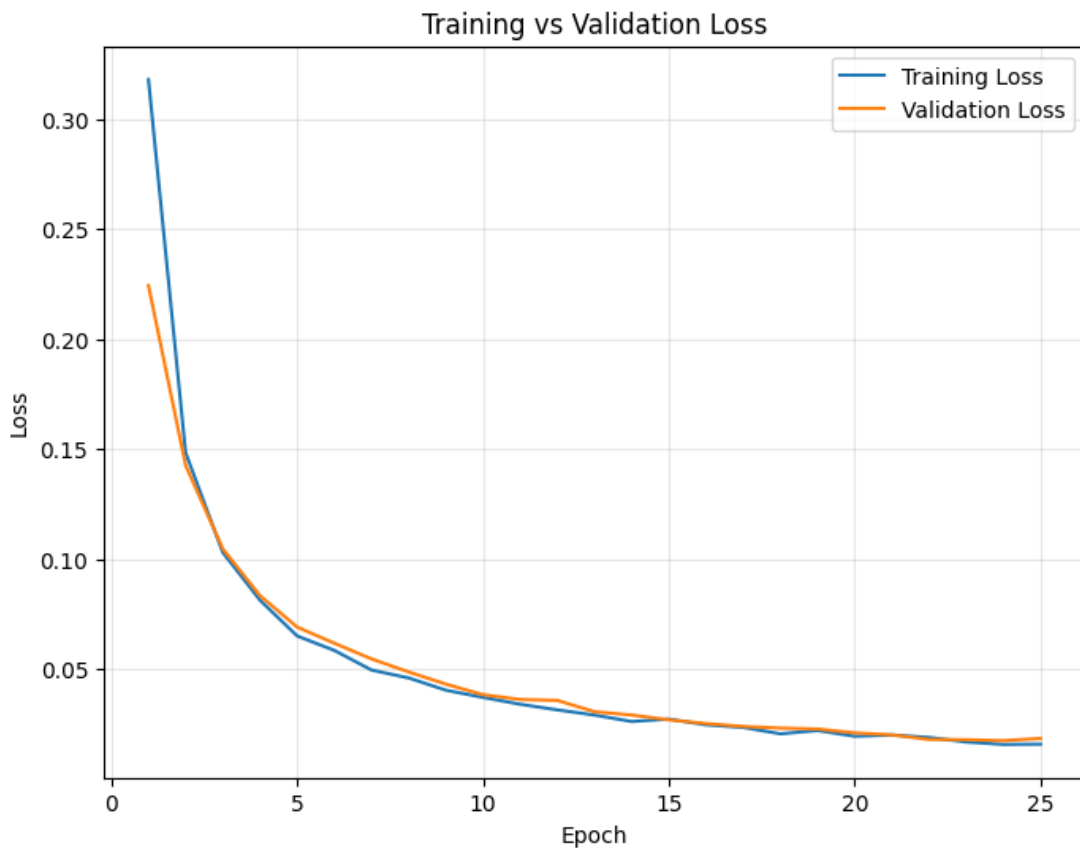


Figure 10: Binary Classification Loss Curve

Figure 7 presents the corresponding training and validation loss curves for Model 1. At epoch 1, the training loss was recorded at approximately 0.32, while the validation loss began at approximately 0.225. Both curves declined sharply within the first two to three epochs, falling to approximately 0.10 by epoch 3, which corresponds directly to the steep accuracy improvement observed during the same period.

Following epoch 5, the decline in loss continued at a more gradual and steady rate, ultimately converging to approximately 0.015–0.02 by epoch 25. A small, temporary increase in validation loss was observed around epoch 12, coinciding precisely with the minor validation accuracy dip noted in Section 4.1.1, further confirming that this fluctuation was a minor and non-recurring anomaly.

Importantly, the training and validation loss curves decreased in parallel throughout training without diverging at any stage. The absence of a rising validation loss trend, despite continued reduction in training loss, provides strong confirmation that the model converged in a stable manner without overfitting.

4.1.3 Confusion Matrix Analysis

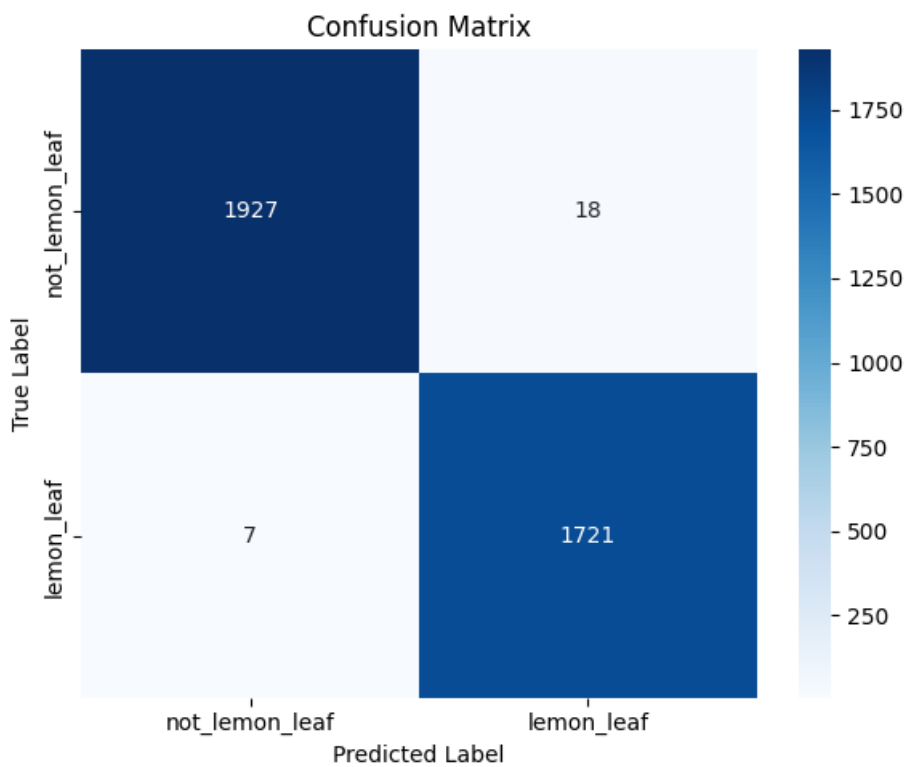


Figure 8: Binary Classification Confusion Matrix

Figure 8 presents the confusion matrix generated from the evaluation of Model 1 on the test dataset, comparing the predicted classification labels against the true labels for the two classes: not_lemon_leaf and lemon_leaf.

As shown in , out of a figure 8 total of 3,673 test images, 1,927 non-lemon-leaf images were correctly classified, and 1,721 lemon-leaf images were correctly

classified, resulting in a total of 3,648 correct predictions. This corresponds to an overall test accuracy of 99.32%.

Of the 25 total misclassifications, 18 were false positives, where non-lemon-leaf images were incorrectly predicted as lemon-leaf images, and 7 were false negatives, where actual lemon-leaf images were incorrectly predicted as not being lemon leaves.

The false-negative rate is of particular operational significance to the overall system, since, according to the System Sequence Diagram presented in Section 3.2.2, an image classified as `not_lemon_leaf` is immediately rejected by the pipeline and does not proceed to the disease classification stage (Model 2). Consequently, any lemon-leaf image incorrectly rejected at this stage would prevent the farmer from receiving a disease diagnosis entirely. The low false-negative count of 7 (out of 1,728 true lemon-leaf test samples) therefore reflects a high degree of reliability in this critical filtering stage. In comparison, the 18 false positives are considered less operationally severe, as such images would proceed to Model 2 but would likely be identified through low-confidence or ambiguous disease predictions rather than resulting in an incorrect treatment recommendation.

4.1.4 Summary

The accuracy and loss curves, together with the confusion matrix, collectively demonstrate that Model 1 achieved stable convergence, strong generalization, and high classification reliability. The close alignment between training and validation performance across both accuracy and loss metrics confirms the absence of overfitting, while the confusion matrix validates the model's practical suitability as the first-stage verification gate within the overall lemon leaf disease detection pipeline.

4.2 Model 2: Multi-Class Disease Classification

4.2.1 Accuracy Curve Analysis

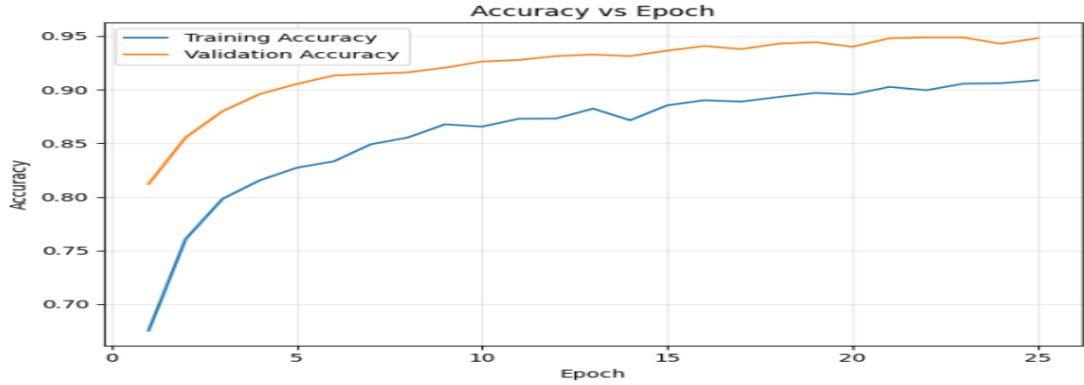


Figure 12: Disease Classification accuracy Curve

Figure 9 presents the training and validation accuracy of the multi-class disease classification model across 25 training epochs. At epoch 1, the training accuracy began at approximately 0.675, while the validation accuracy started at a comparatively higher value of approximately 0.81. Both curves increased steadily throughout training; however, the validation accuracy rose more rapidly than the training accuracy during the early epochs, reaching approximately 0.93 by epoch 10, while training accuracy reached only approximately 0.87 at the same point.

From epoch 15 onward, both curves entered a plateau phase, with validation accuracy stabilizing in the range of 0.945–0.95, and training accuracy stabilizing at approximately 0.90–0.91. A minor fluctuation is visible in validation accuracy around epoch 24, where it dips slightly before recovering to approximately 0.947 by epoch 25. Training accuracy concluded at approximately 0.91.

Unlike Model 1, validation accuracy remained consistently higher than training accuracy throughout the entire training process. This behavior is attributed to the Dropout and L2 regularization applied to Model 2, as specified in Section 3.3 (Model Training). Dropout is active only during the training phase, where it randomly deactivates a portion of neurons to prevent overfitting, thereby lowering the training accuracy relative to its true underlying capability. During validation and testing, dropout is deactivated, allowing the full network to operate, which results in a higher

measured accuracy. This is a well-understood and expected characteristic of regularized deep learning models rather than an indication of data leakage or evaluation error.

4.2.2 Loss Curve Analysis

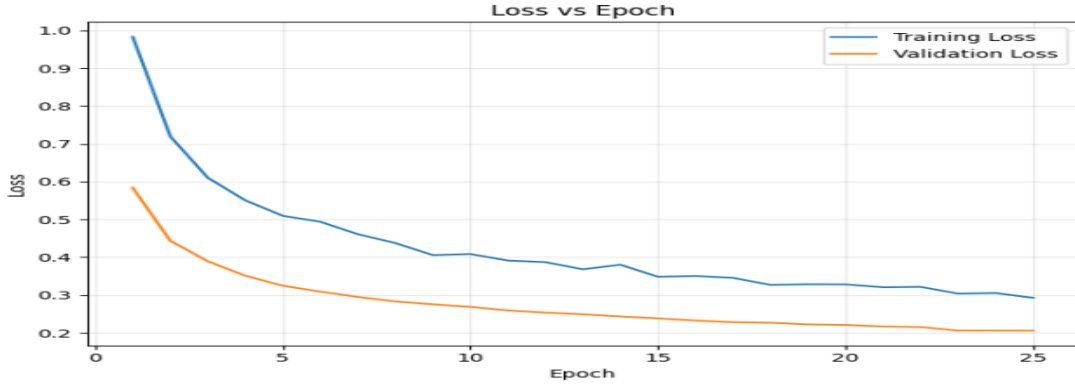


Figure 13: Disease Classification Loss Curve

Figure 10 presents the corresponding training and validation loss curves for Model 2. At epoch 1, training loss began at approximately 0.99, while validation loss began at approximately 0.59. Both curves declined rapidly during the first 10 epochs, with training loss falling to approximately 0.40 and validation loss falling to approximately 0.27.

Beyond epoch 10, the rate of decline slowed considerably, and both curves continued to decrease gradually and smoothly until epoch 25, at which point training loss reached approximately 0.30 and validation loss reached approximately 0.21. Consistent with the accuracy curve, validation loss remained lower than training loss throughout training, for the same regularization-related reason discussed in Section 4.2.1.

Both curves flattened smoothly after approximately epoch 15, with no divergence or upward trend observed in the validation loss. This confirms that, despite the increased complexity of the nine-class classification task relative to Model 1's binary task, the model converged in a stable manner without overfitting.

4.2.3 Confusion Matrix Analysis

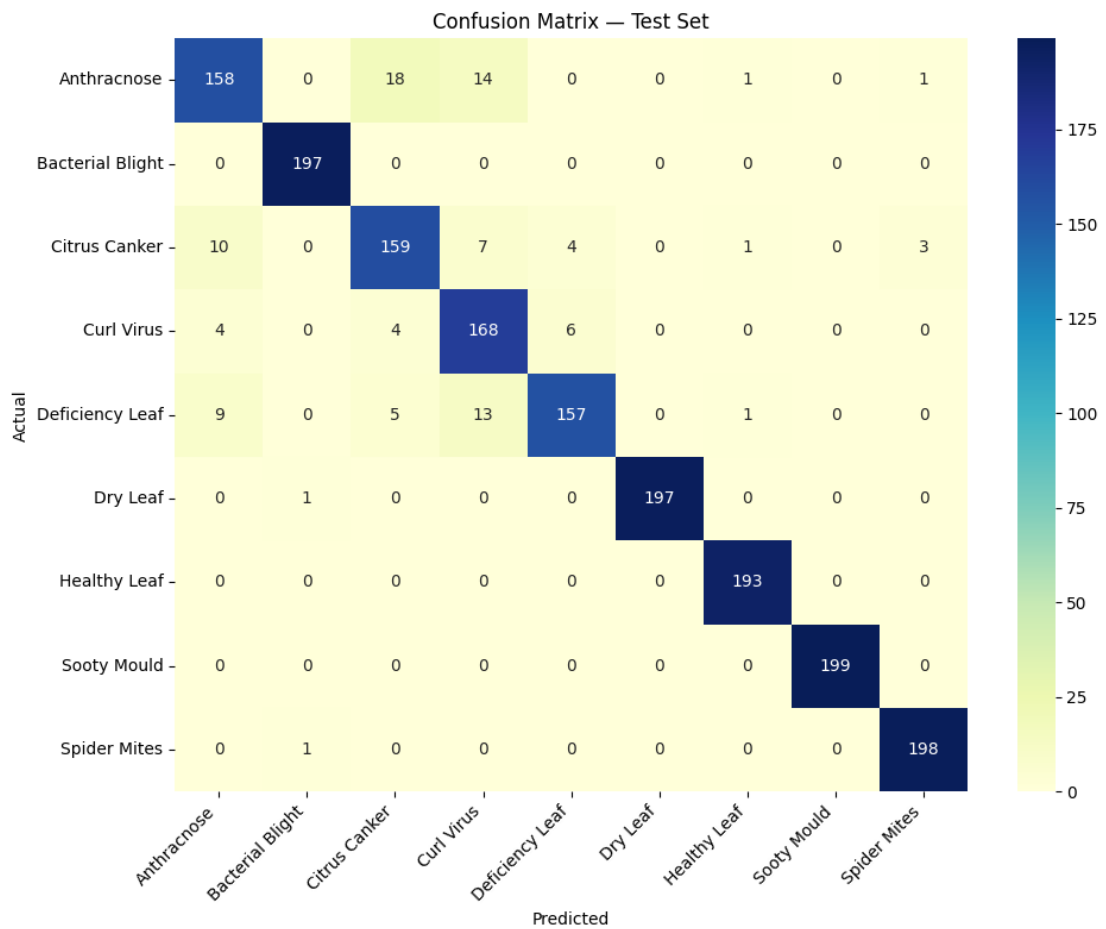


Figure 14: Disease Classification Confusion Matrix

Figure 11 presents the confusion matrix generated from the evaluation of Model 2 on the test dataset across all nine disease classes: Anthracnose, Bacterial Blight, Citrus Canker, Curl Virus, Deficiency Leaf, Dry Leaf, Healthy Leaf, Sooty Mould, and Spider Mites.

As shown in Figure 11 , out of a total of 1,729 test images, 1,626 were correctly classified, resulting in an overall test accuracy of 94.04%, consistent with the value reported in Section 3.3 (Evaluation).

Class-wise analysis of the confusion matrix reveals two distinct performance groups. The first group, comprising Bacterial Blight, Healthy Leaf, and Sooty Mould, achieved perfect classification (100%), while Dry Leaf and Spider Mites achieved near-perfect accuracy (99.5%). These classes possess visually distinctive symptoms such as the uniform dark coating of sooty mould or the visible webbing associated

with spider mite infestation which allow the convolutional layers to extract highly discriminative features with minimal ambiguity.

The second group, comprising Anthracnose, Citrus Canker, Curl Virus, and Deficiency Leaf, exhibited comparatively lower class-wise accuracy, ranging from 82.3% to 92.3%, and accounted for nearly all of the model's misclassifications. Specific error patterns observed in the confusion matrix include:

- 18 Anthracnose samples misclassified as Citrus Canker, and 14 misclassified as Curl Virus, together accounting for the majority of Anthracnose misclassifications.
- 13 Deficiency Leaf samples misclassified as Curl Virus.
- 10 Citrus Canker samples misclassified as Anthracnose, indicating a reciprocal confusion pattern between these two classes.

Notably, no misclassification occurred between the four confused classes and the five well-separated classes, indicating that the confusion is confined specifically to visually and symptomatically similar disease categories, likely arising from overlapping characteristics such as leaf discoloration patterns, lesion shape, and spot density.

4.2.4 Summary

The accuracy and loss curves for Model 2 confirm stable convergence and effective generalization despite the increased complexity of classifying nine distinct disease categories. The confusion matrix further demonstrates that the model performs with near-perfect reliability on visually distinctive disease classes, while classification errors remain concentrated among four biologically and visually similar disease categories. These findings, achieving an overall accuracy of 94.04%, indicate that Model 2 provides a reliable and practically deployable solution for multi-class lemon leaf disease classification, while also identifying a clear direction for future model refinement.

CHAPTER:5

CONCLUSION

5.1 Conclusion

This project successfully developed an automated lemon leaf disease detection system using a two-stage CNN pipeline built on MobileNetV2. The first stage, a binary verification model, achieved 99.32% accuracy in confirming whether an uploaded image is a lemon leaf, while the second stage, a multi-class disease classification model, achieved 94.04% accuracy across nine disease categories: Anthracnose, Bacterial Blight, Citrus Canker, Curl Virus, Deficiency Leaf, Dry Leaf, Healthy Leaf, Sooty Mould, and Spider Mites. Five of the nine classes were classified with near-perfect accuracy (99.5–100%), while the remaining four visually similar classes showed comparatively higher confusion, consistent with patterns observed in related literature. The stable training and validation curves for both models confirmed effective convergence without overfitting, validating the preprocessing, augmentation, and regularization strategies used. By expanding classification to nine classes on a combined dataset of 8,644 images, this project directly addressed the research gap identified in Chapter 2, where prior studies were largely limited to two-to-four disease categories. Overall, the project delivers a fast, accurate, and farmer-accessible tool that supports early disease detection, reduces reliance on expert consultation, and contributes to precision agriculture and sustainable lemon farming practices.

5.2 Limitations

- Reduced accuracy (82.3–92.3%) for four visually similar disease classes (Anthracnose, Citrus Canker, Curl Virus, Deficiency Leaf).
- Limited validation under real-world field conditions (lighting, blur, occlusion).
- Does not account for co-occurring diseases or early-stage mild symptoms.
- Generalization to new regions or lemon varieties not yet field-tested.

5.3 Future Work

Building on the outcomes of this project, the following directions are recommended for future development:

- **Real-world field validation:** Deploying and testing the system with images captured directly by farmers under diverse field conditions to evaluate real-world robustness and reliability.
- **Expanded disease coverage:** Incorporating additional disease classes, including citrus greening (Huanglongbing) and citrus leaf miner damage, which were present in the supplementary dataset but excluded from the final class selection, to provide more comprehensive diagnostic coverage.
- **Improved discrimination among similar classes:** Exploring higher input resolutions, attention-based mechanisms, or fine-grained feature extraction techniques to reduce confusion among visually similar disease categories such as Anthracnose, Citrus Canker, Curl Virus, and Deficiency Leaf.

BIBLIOGRAPHY

(n.d.). Retrieved from

<https://journals.plos.org/plosone/article/figures?id=10.1371/journal.pone.0282336>

(n.d.). Retrieved from https://www.researchgate.net/figure/General-Architecture-of-CNN_fig2_351172173

Aktaruzzaman Pramanik, A. Z. (2021, 07). Retrieved from
10.1109/ICCCNT51525.2021.9579586

Arifin, K. N., & Jahan, I. (2024). Retrieved from
<https://doi.org/10.1145/3723178.3723199>

Benjamin Doh (Jiangsu University, C. (n.d.). *IEEE*. Retrieved from <https://iee-dataport.org/documents/lemon-leaf-disease-dataset>

da Silva, J. C., & Oliveira, R. A. (2023). Retrieved from
<https://doi.org/10.3390/s23042165>

Dhiman, P. K., & Ababneh, N. (2023). Retrieved from
<https://doi.org/10.3390/su15054576>

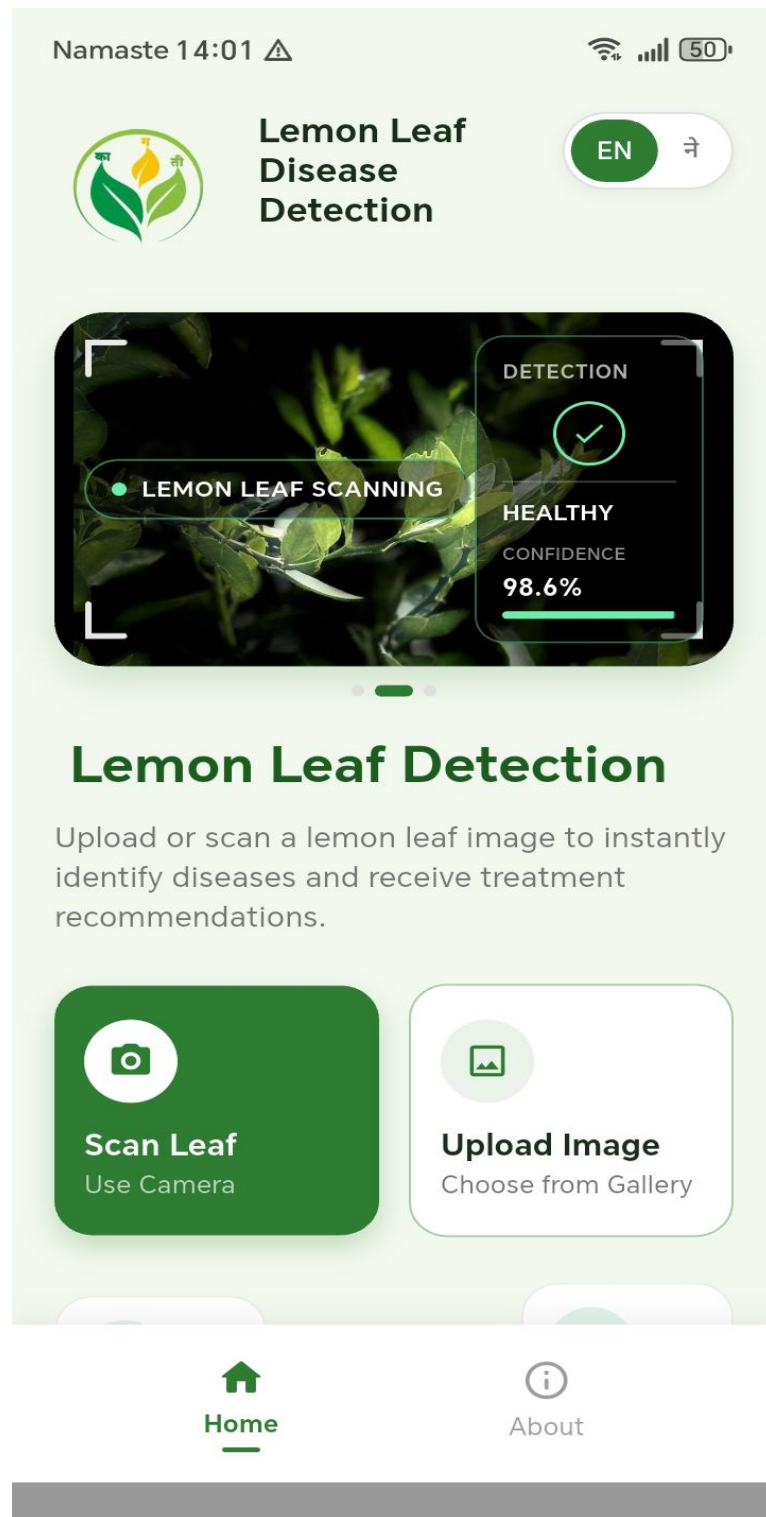
Diagram, S. S.

Kalejahi, B. K. (2026). *Lightweight Deep Learning Models for Early Plant Disease Detection*. Retrieved from
https://jpis.az/uploads/article/en/2026_1/LIGHTWEIGHT_DEEP_LEARNING_G_MODELS_FOR_EARLY_PLANT_DISEASE_DETECTION_A_TRANSFER_LEARNING_APPROACH_WITH_MOBILENETV2_AND_RESNET50.pdf

Md Naimul Hasan, M. F. (2025). Retrieved from
<https://data.mendeley.com/datasets/smz8ffbthy/3>

Saygılı, A. (2025). Retrieved from <https://doi.org/10.3390/app152010988>
UseCase Diagram.

APPENDICES



Appendix1: Home page

9 Lemon Leaf Classes Detected

Our CNN model is trained to classify lemon leaf images into one healthy category and eight common disease, pest, or leaf-condition categories.



Healthy

Healthy

Healthy lemon leaves are generally green, smooth, and free from maj...

[View Details >](#)



Anthracnose

Fungal Disease

Anthracnose is a fungal disease that may cause dark brown or bla...

[View Details >](#)



Bacterial Blight

Bacterial Disease

Bacterial blight can cause water-soaked spots, dark lesions, yellowi...

[View Details >](#)



Citrus Canker

Bacterial Disease

Citrus canker is a bacterial disease characterized by raised, rough, co...

[View Details >](#)



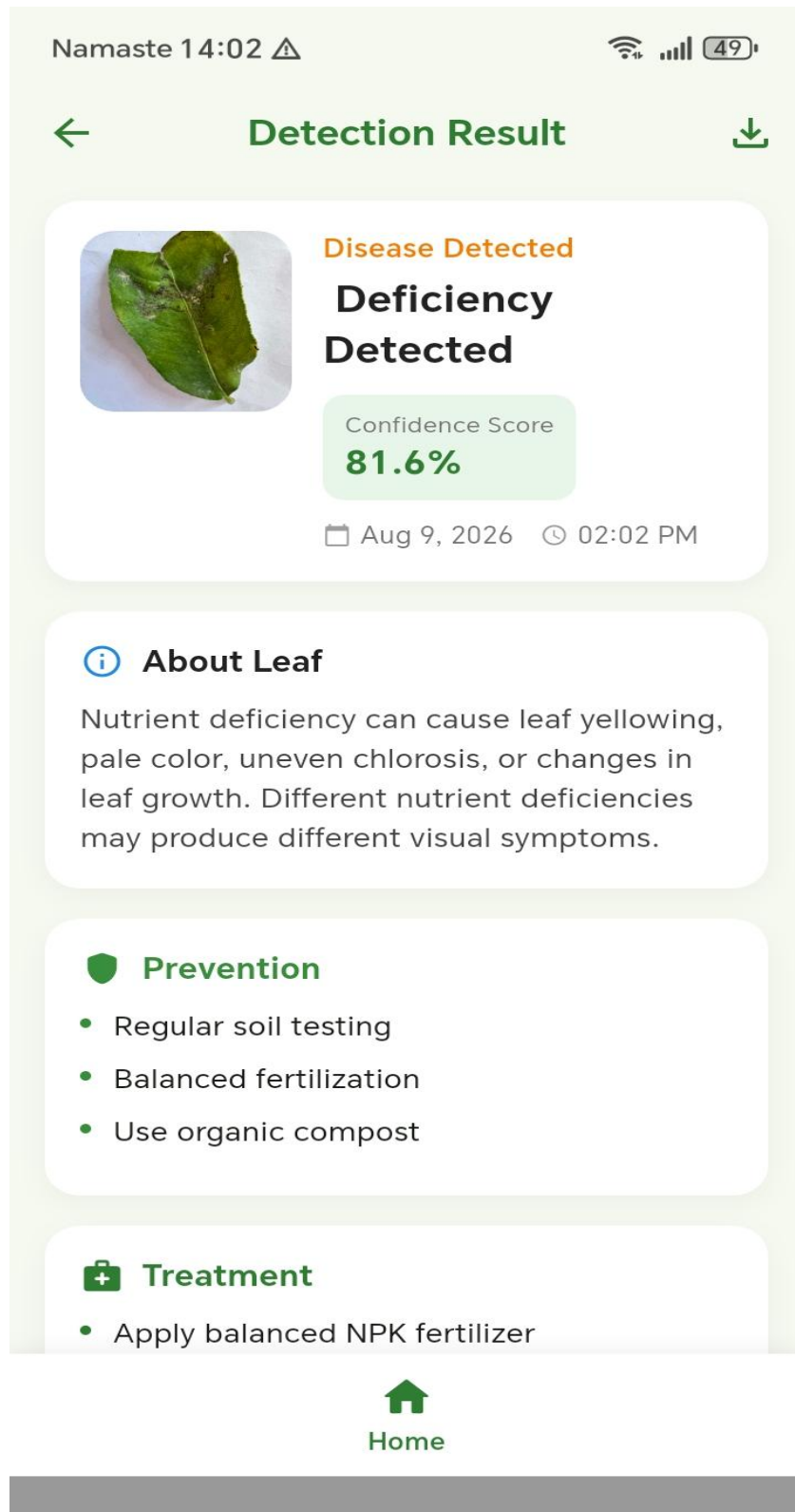
Curl Virus

Viral Disease

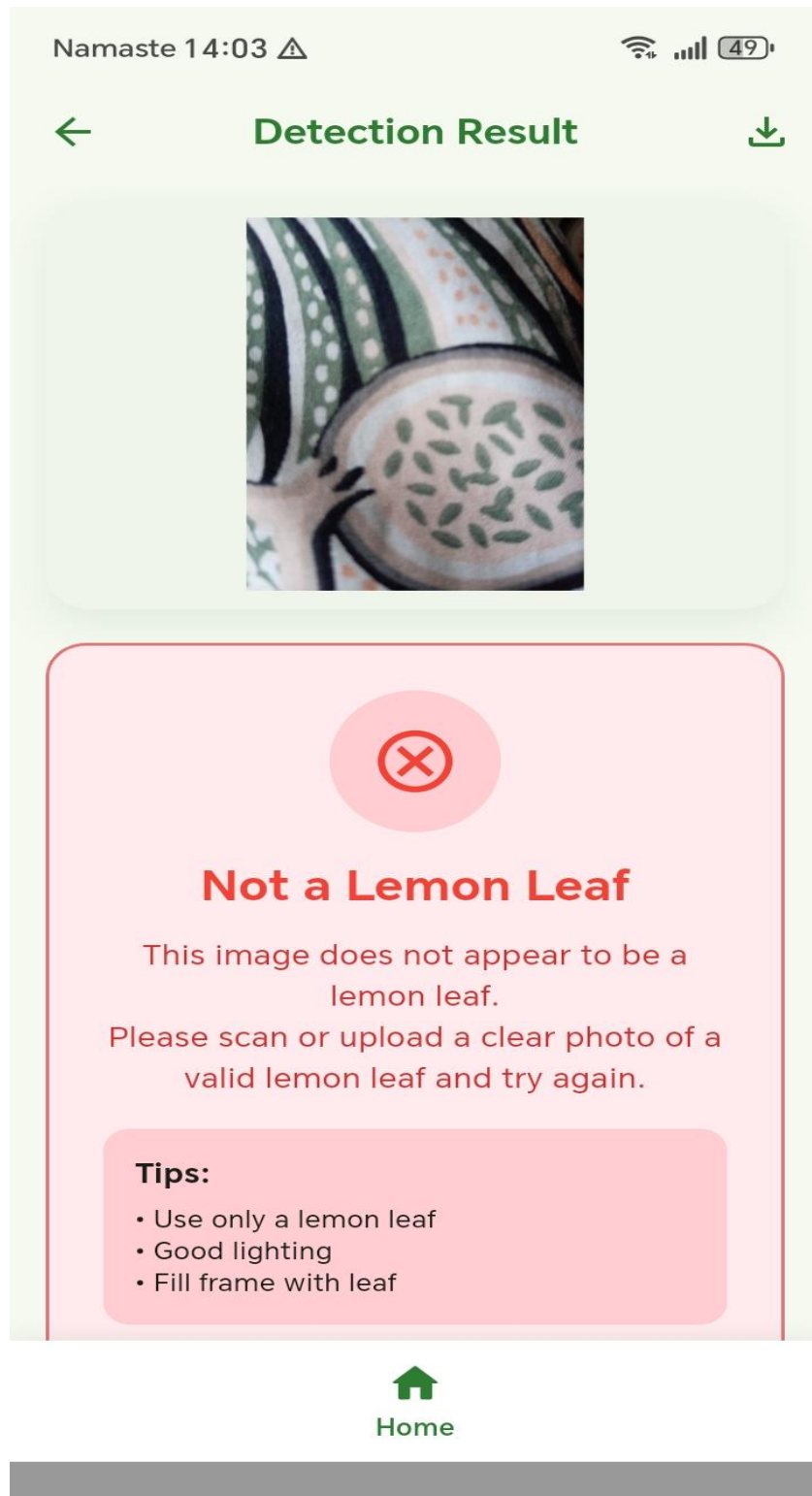
Curl virus infection may cause leaves to curl, twist, become disto...

[View Details >](#)

Apendix2: About page



Appendix3: Lemon Leaf Disease Detection Result



Appendix4: Non Lemon Leaf Detection Result